

Lab 1.1 – Using Wireshark to View Network Traffic

Student Name: Mohamad Mostafa (ID: 30930)

Part 1: Capture and Analyze Local ICMP Data in Wireshark

Step 1: Retrieve your PC Interface Addresses

IPv4 Address: 192.168.1.104

MAC Address: 9C-AD-97-30-57-CF

NIC Description: Ralink RT3290 802.11bgn Wi-Fi Adapter

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Wireless LAN adapter Wi-Fi 4:
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Connection-specific DNS Suffix . : 
Description . . . . . : Ralink RT3290 802.11bgn Wi-Fi Adapter
Physical Address. . . . . : 9C-AD-97-30-57-CF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::698:5bd5:b97b:ddfc%28(Preferred)
IPv4 Address. . . . . : 192.168.1.104(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, March 4, 2026 1:26:13 PM
Lease Expires . . . . . : Saturday, March 7, 2026 1:26:12 PM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 446475671
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-9E-0C-79-5C-60-BA-B5-99-DA
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

Other device IPv4 Address: 192.168.1.101

Step 2: Start Wireshark and Begin Capturing Data

Wi-Fi 4 interface (promiscuous mode) is enabled.

	Interface	Traffic	Link-layer Header	Promiscuous
	Local Area Connection* 8	_____	Ethernet	✓
	Local Area Connection* 7	_____	Ethernet	✓
	Local Area Connection* 6	_____	Ethernet	✓
▶	Wi-Fi 4		Ethernet	✓
▶	Local Area Connection* 12	_____	Ethernet	✓
▶	Local Area Connection* 11	_____	Ethernet	✓
▶	vEthernet (WSL)	_____	Ethernet	✓
▶	Adapter for loopback traffic capture	_____	BSD loopback	✓
▶	Ethernet 6	_____	Ethernet	✓

The comment matches my NIC description -> Ralink RT3290 802.11bgn Wi-Fi Adapter.

Manage Interfaces			
Local Interfaces			
Show	Friendly Name	Interface Name	Comment
✓	Local Area Con...	\Device\NPF_{E7800935-E23A-43BF-B545-3E7528C51DEF}	WAN Miniport (Network Monitor)
✓	Local Area Con...	\Device\NPF_{B84DCBEA-B385-4661-BEC0-4134C2920B37}	WAN Miniport (IPv6)
✓	Local Area Con...	\Device\NPF_{A8748416-A2B0-40DF-940B-2E74E24FF554}	WAN Miniport (IP)
✓	Wi-Fi 4	\Device\NPF_{DD6F17D4-562D-4C91-825D-CF59EEEE91D7}	Ralink RT3290 802.11bgn Wi-Fi Adapter
✓	Local Area Con...	\Device\NPF_{A37B3C5F-8E5C-4A48-8190-6EBE5325712A}	Microsoft Wi-Fi Direct Virtual Adapter #4
✓	Local Area Con...	\Device\NPF_{F5F686F0-A8C2-4479-980D-559A2CA3AF24}	Microsoft Wi-Fi Direct Virtual Adapter #3
✓	vEthernet (WSL)	\Device\NPF_{295BD63D-F996-417A-B142-3BA848F4315B}	Hyper-V Virtual Ethernet Adapter
✓	Adapter for loo...	\Device\NPF_Loopback	
✓	Ethernet 6	\Device\NPF_{1ED13323-CE8A-40A5-87C2-538EF8961225}	Realtek PCIe FE Family Controller

Wireshark started.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.104	192.168.0.27	TCP	66	63316 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2	1.006565	192.168.1.104	192.168.0.27	TCP	66	[TCP Retransmission] 63316 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
3	3.007300	192.168.1.104	192.168.0.27	TCP	66	[TCP Retransmission] 63316 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
4	3.277045	192.168.1.1	192.168.1.255	RIPv2	86	Response
5	3.375901	192.168.1.102	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
6	3.679130	192.168.1.108	192.168.1.255	UDP	62	59363 → 7721 Len=20
7	3.780968	192.168.1.1	239.255.255.250	SSDP	308	NOTIFY * HTTP/1.1
8	3.883823	192.168.1.1	239.255.255.250	SSDP	308	NOTIFY * HTTP/1.1
9	4.090193	192.168.1.1	239.255.255.250	SSDP	356	NOTIFY * HTTP/1.1
10	4.297171	192.168.1.1	239.255.255.250	SSDP	376	NOTIFY * HTTP/1.1
11	4.906630	fe80::8c66:6bff::	ff02::fb	MDNS	123	Standard query 0x0016 PTR _googlecast._tcp.local, "QM" question PTR _233637DE._sub._goog
12	4.908703	192.168.1.1	239.255.255.250	SSDP	372	NOTIFY * HTTP/1.1
13	5.010324	192.168.1.1	239.255.255.250	SSDP	372	NOTIFY * HTTP/1.1
14	5.112441	192.168.1.1	239.255.255.250	SSDP	356	NOTIFY * HTTP/1.1

Pinging 192.168.1.101

```
C:\Users\10User>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:
Reply from 192.168.1.101: bytes=32 time=449ms TTL=64
Reply from 192.168.1.101: bytes=32 time=434ms TTL=64
Reply from 192.168.1.101: bytes=32 time=131ms TTL=64
Reply from 192.168.1.101: bytes=32 time=130ms TTL=64

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 130ms, Maximum = 449ms, Average = 286ms
```

Capturing ICMP data in Wireshark :

icmp							
No.	Time	Source	Destination	Protocol	Length	Info	
→	395.254.036811	192.168.1.104	192.168.1.101	ICMP	74	Echo (ping) request	id=0x0001
←	396.254.166838	192.168.1.101	192.168.1.104	ICMP	74	Echo (ping) reply	id=0x0001
	398.254.741730	192.168.1.104	192.168.1.101	ICMP	74	Echo (ping) request	id=0x0001
	399.255.175782	192.168.1.101	192.168.1.104	ICMP	74	Echo (ping) reply	id=0x0001
	403.255.766429	192.168.1.104	192.168.1.101	ICMP	74	Echo (ping) request	id=0x0001
	404.255.897390	192.168.1.101	192.168.1.104	ICMP	74	Echo (ping) reply	id=0x0001
	406.256.790601	192.168.1.104	192.168.1.101	ICMP	74	Echo (ping) request	id=0x0001
	407.256.921067	192.168.1.101	192.168.1.104	ICMP	74	Echo (ping) reply	id=0x0001

Step 3: Examine the Captured Data

Clicking the first ICMP (ping) request. (MY PC -> OTHER DEVICE)

Source IP@: 192.168.1.104

-> Destination IP@: 192.168.1.101

Source MAC@: 9C-AD-97-30-57-CF

-> Destination MAC@: C0-BD-C8-F7-18-3D

```
▶ Frame 395: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \De
▼ Ethernet II, Src: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf), Dst: SamsungElect_f7:18:3d (c0:b
  ▶ Destination: SamsungElect_f7:18:3d (c0:bd:c8:f7:18:3d)
  ▶ Source: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
    Type: IPv4 (0x0800)
    [Stream index: 26]
  ▶ Internet Protocol Version 4, Src: 192.168.1.104, Dst: 192.168.1.101
  ▶ Internet Control Message Protocol
```

My PC obtains the MAC address of the PC I am ping using a protocol called ARP (Address Resolution Protocol).

ARP	42 Who has 192.168.1.101? Tell 192.168.1.
ARP	42 Who has 192.168.1.101? Tell 192.168.1.
ARP	42 192.168.1.101 is at c0:bd:c8:f7:18:3d
ARP	42 192.168.1.101 is at c0:bd:c8:f7:18:3d
ARP	42 192.168.1.101 is at c0:bd:c8:f7:18:3d

Encapsulation Analysis:

When generating a ping, the ICMP protocol first wraps the user payload (like a timestamp or text string) with an ICMP header containing type, code, and checksum to create an ICMP segment. This segment is then passed to the Internet Protocol, which adds an IP header containing the source and destination IP addresses, protocol field (indicating ICMP), and TTL, forming an IP packet. Finally, the Data Link layer encapsulates this entire IP packet within an Ethernet frame, adding a header with source and destination MAC addresses (determined via ARP) and an EtherType field, plus a trailer with a Frame Check Sequence for error detection. Once fully encapsulated, the Network Interface Card converts this frame into electrical signals, light pulses, or radio waves for transmission across the physical LAN medium.

Part 2: Capture and Analyze Remote ICMP Data in Wireshark

Step 1: Start Capturing Data on the Interface

Pinging google.com

```
C:\Users\10User>ping google.com

Pinging google.com [142.251.36.46] with 32 bytes of data:
Reply from 142.251.36.46: bytes=32 time=83ms TTL=114
Reply from 142.251.36.46: bytes=32 time=79ms TTL=114
Reply from 142.251.36.46: bytes=32 time=79ms TTL=114
Reply from 142.251.36.46: bytes=32 time=82ms TTL=114

Ping statistics for 142.251.36.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 79ms, Maximum = 83ms, Average = 80ms
```

Destination IP Address: 142.251.36.46

icmp						
No.	Time	Source	Destination	Protocol	Length	Info
→	20 7.138265	192.168.1.115	142.251.36.46	ICMP	74	Echo (ping) request
←	21 7.221542	142.251.36.46	192.168.1.115	ICMP	74	Echo (ping) reply
	23 8.152541	192.168.1.115	142.251.36.46	ICMP	74	Echo (ping) request
	24 8.231835	142.251.36.46	192.168.1.115	ICMP	74	Echo (ping) reply
	25 9.173776	192.168.1.115	142.251.36.46	ICMP	74	Echo (ping) request
	26 9.253592	142.251.36.46	192.168.1.115	ICMP	74	Echo (ping) reply
	27 10.187804	192.168.1.115	142.251.36.46	ICMP	74	Echo (ping) request
	28 10.269668	142.251.36.46	192.168.1.115	ICMP	74	Echo (ping) reply

Destination MAC Address: c0:06:c3:2c:c5:71

```
▶ Frame 20: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{...}
▼ Ethernet II, Src: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf), Dst: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
    ▶ Destination: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
    ▶ Source: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
    Type: IPv4 (0x0800)
    [Stream index: 2]
▶ Internet Protocol Version 4, Src: 192.168.1.115, Dst: 142.251.36.46
▶ Internet Control Message Protocol
```

Pinging cisco.com

```
C:\Users\10User>ping www.cisco.com

Pinging e2867.dsca.akamaiedge.net [2.22.4.110] with 32 bytes of data:
Reply from 2.22.4.110: bytes=32 time=107ms TTL=50
Reply from 2.22.4.110: bytes=32 time=87ms TTL=50
Reply from 2.22.4.110: bytes=32 time=82ms TTL=50
Reply from 2.22.4.110: bytes=32 time=85ms TTL=50

Ping statistics for 2.22.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 82ms, Maximum = 107ms, Average = 90ms
```

IP Address: 2.22.4.110

icmp && ip.addr==2.22.4.110							
No.	Time	Source	Destination	Protocol	Length	Info	
→ 1445	344.852083	192.168.1.104	2.22.4.110	ICMP	74	Echo (ping)	request
← 1447	344.959325	2.22.4.110	192.168.1.104	ICMP	74	Echo (ping)	reply
1460	345.870606	192.168.1.104	2.22.4.110	ICMP	74	Echo (ping)	request
1470	345.957449	2.22.4.110	192.168.1.104	ICMP	74	Echo (ping)	reply
1480	346.899973	192.168.1.104	2.22.4.110	ICMP	74	Echo (ping)	request
1481	346.982580	2.22.4.110	192.168.1.104	ICMP	74	Echo (ping)	reply
1483	347.912123	192.168.1.104	2.22.4.110	ICMP	74	Echo (ping)	request
1484	347.997786	2.22.4.110	192.168.1.104	ICMP	74	Echo (ping)	reply

Destination MAC Address: c0:06:c3:2c:c5:71

```
▶ Frame 1445: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \De
▼ Ethernet II, Src: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf), Dst: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
  ▶ Destination: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
    ▼ Source: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
      .... ..0. .... = LG bit: Globally unique address (factory default)
      .... ..0. .... = IG bit: Individual address (unicast)
      Type: IPv4 (0x0800)
      [Stream index: 2]
    ▶ Internet Protocol Version 4, Src: 192.168.1.104, Dst: 2.22.4.110
    ▶ Internet Control Message Protocol
```

Pinging yahoo.com

```
C:\Users\10User>ping yahoo.com

Pinging yahoo.com [98.137.11.164] with 32 bytes of data:
Reply from 98.137.11.164: bytes=32 time=250ms TTL=43
Reply from 98.137.11.164: bytes=32 time=243ms TTL=43
Reply from 98.137.11.164: bytes=32 time=245ms TTL=43
Reply from 98.137.11.164: bytes=32 time=243ms TTL=43

Ping statistics for 98.137.11.164:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 243ms, Maximum = 250ms, Average = 245ms
```

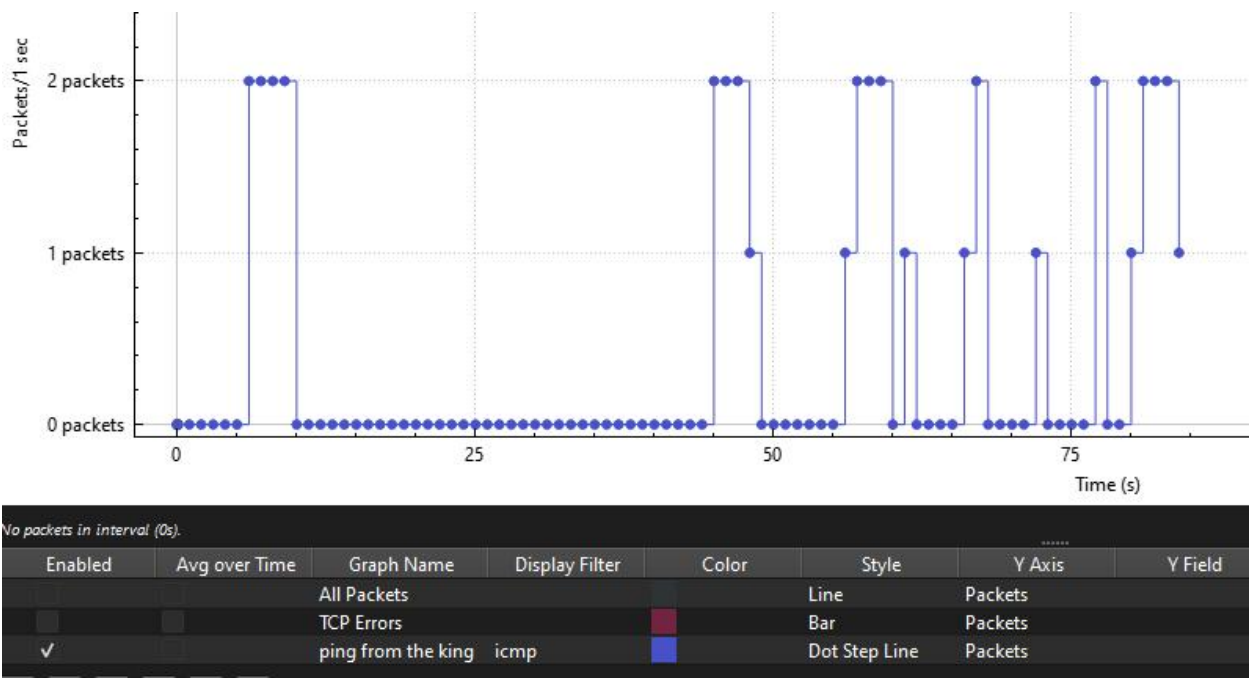
Destination IP Address: 98.137.11.164

icmp && ip.addr==98.137.11.164						
No.	Time	Source	Destination	Protocol	Length	Info
→	5151 684.523561	192.168.1.104	98.137.11.164	ICMP	74	Echo (ping) request
←	5152 684.773631	98.137.11.164	192.168.1.104	ICMP	74	Echo (ping) reply
	5153 685.556698	192.168.1.104	98.137.11.164	ICMP	74	Echo (ping) request
	5154 685.800369	98.137.11.164	192.168.1.104	ICMP	74	Echo (ping) reply
	5155 686.573487	192.168.1.104	98.137.11.164	ICMP	74	Echo (ping) request
	5156 686.818843	98.137.11.164	192.168.1.104	ICMP	74	Echo (ping) reply
	5157 687.604982	192.168.1.104	98.137.11.164	ICMP	74	Echo (ping) request
	5158 687.848318	98.137.11.164	192.168.1.104	ICMP	74	Echo (ping) reply

Destination MAC Address: c0:06:c3:2c:c5:71

```
► Frame 5151: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF...
▼ Ethernet II, Src: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf), Dst: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
  Destination: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 2]
```

ICMP Graph Statistics (Packets/seconds)



Step 2: Examine and Analyze the Data from Remote Hosts

Comparing MAC@ of the three remote locations :

They are all identical, and they are all the MAC address of my router (Default Gateway).

The frame is addressed to the router's MAC, while the packet inside is addressed to the remote PC's IP.

Pinging my default gateway (192.168.1.1)

icmp && ip.addr==192.168.1.1						
No.	Time	Source	Destination	Protocol	Length	Info
→ 6237	885.552354	192.168.1.104	192.168.1.1	ICMP	74	Echo (ping) request
← 6238	885.554215	192.168.1.1	192.168.1.104	ICMP	74	Echo (ping) reply
→ 6240	886.564567	192.168.1.104	192.168.1.1	ICMP	74	Echo (ping) request
→ 6241	886.567607	192.168.1.1	192.168.1.104	ICMP	74	Echo (ping) reply
→ 6247	887.591486	192.168.1.104	192.168.1.1	ICMP	74	Echo (ping) request
→ 6248	887.598017	192.168.1.1	192.168.1.104	ICMP	74	Echo (ping) reply
→ 6249	887.598737	192.168.1.1	192.168.1.104	ICMP	74	Echo (ping) reply
→ 6250	888.623692	192.168.1.104	192.168.1.1	ICMP	74	Echo (ping) request
→ 6251	888.625173	192.168.1.1	192.168.1.104	ICMP	74	Echo (ping) reply

Destination MAC Address: c0:06:c3:2c:c5:71

▶ Frame 6237: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \De
▼ Ethernet II, Src: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf), Dst: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
▼ Destination: TPLink_2c:c5:71 (c0:06:c3:2c:c5:71)
.... ..0. = LG bit: Globally unique address (factory default)
.... ..0. = IG bit: Individual address (unicast)
▼ Source: HonHaiPrecis_30:57:cf (9c:ad:97:30:57:cf)
.... ..0. = LG bit: Globally unique address (factory default)
.... ..0. = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)

Local Ping: The Destination MAC address is the Target Device's MAC. They are in the same LAN, so they can talk directly. My PC sends an ARP broadcast asking "Who has this Target IP?" The Target PC replies.

Remote Ping: The Destination MAC address is the Router's MAC (Default Gateway). My PC needs to send the frame to the router to get out of the local network. My PC sends an ARP broadcast asking "Who has my Gateway IP?" The Router replies.

Key Concepts Summary

ICMP (Internet Control Message Protocol)

Used by ping to test connectivity. Echo Requests are sent; Echo Replies confirm reachability.

Wireshark Display Filter

Typing 'icmp' in the filter bar shows only ICMP frames, making it easier to examine ping traffic.

MAC Address (Layer 2)

Physical address of the NIC. Obtained via ARP for local hosts. For remote hosts, the destination MAC belongs to the default gateway router, not the final destination.

IP Address (Layer 3)

Logical address used end-to-end. The destination IP is always the final host, even for remote traffic.

ARP (Address Resolution Protocol)

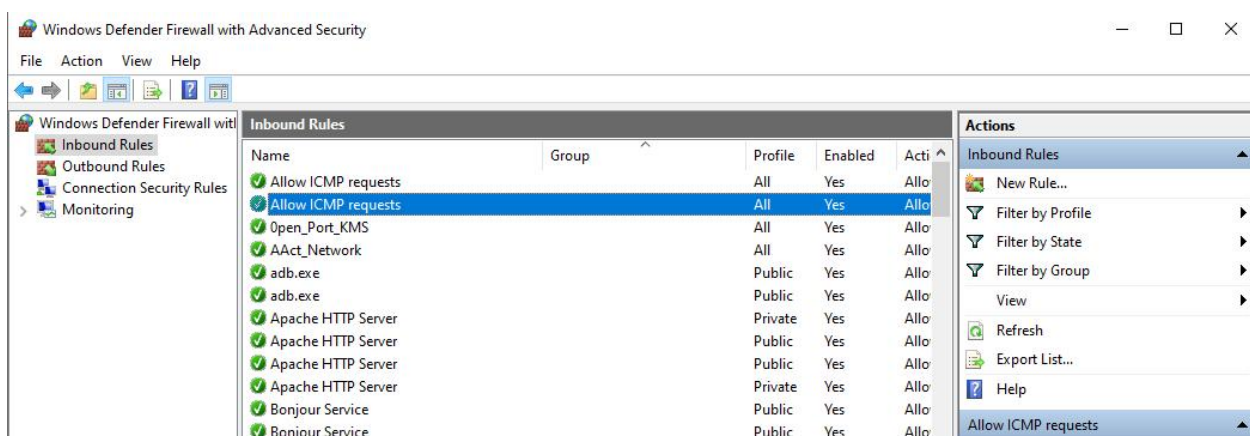
Resolves an IPv4 address to a MAC address within the same LAN segment.

Default Gateway

When the destination is outside the local network, the frame is addressed (at Layer 2) to the default gateway router's MAC address.

Lab Completion Checklist

- ☒ Successfully captured local ICMP traffic
- ☒ Analyzed local MAC and IP addresses
- ☒ Successfully captured remote ICMP traffic to three websites
- ☒ Compared and contrasted local vs. remote packet data
- ☒ Completed all reflection questions
- ☒ Disabled/removed ICMP firewall rule



THANK YOU.

Mohamad Mostafa (ID: 30930)